Elias Pedroza

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Education

Northwestern University — **Expected Dec. 2027**
Master of Science in Robotics — *Evanston, IL*

Santa Clara University — **Jun. 2026**
Bachelor of Science in Mechanical Engineering, cum laude — *Santa Clara, CA*

Experience

1X Technologies — **Mar. 2026 – Present**
Robot Service Engineering Technician — *SF Bay Area, CA*

- Diagnose mechanical, electrical, & software failures via inspection, terminal commands, & Robot Visualizer digital twins
- Perform precision repairs on humanoid robots, interpreting CAD & electrical schematics to verify calibration & wiring
- Documented root causes & executed test sequences to verify system performance, managing full repair life cycles in Jira

Design for Assistive Robotic Technologies (DART) Lab — **Jun. 2025 – Jun. 2026**
DART Lab Research Intern — *Santa Clara, CA*

- Prototyped jamming-based flexures, cutting thickness by 55% while simultaneously improving stiffness control & ergonomics via material & geometric iteration
- Designed CAD assemblies for silicone molds & flexure fingers, enabling seamless integration with the granular jamming palm while supporting versatile grasp orientations
- Developed test rigs to quantify deflection & stiffness of laminar joints, improving measurement precision & repeatability
- Implemented a whiffletree load-sharing mechanism to distribute forces across fingers, improving grasp capability

NASA L'SPACE Mission Concept Academy — **Jan. 2025 – May 2025**
Thermal Engineer — *Remote*

- Designed thermal protection for a Venus-bound Aerobot, selecting insulation, heaters, & cooling systems to maintain component safety between 225.8 K (53 °F) & 344.7 K (161 °F)
- Developed heat flow maps to model the payload steady-state behavior under solar, albedo, convection, & radiation loads
- Presented & co-authored a 190-page design review, coordinating subsystems with a 12-member multidisciplinary team

Projects

Integrated Reef Information System (IRIS) | *SCU Robotics Systems Lab (RSL) (Senior Design)*

- Developing an autonomous buoy–winch system to retrieve data from a submerged sensor network, eliminating diver-based collection & enabling protected submergence between retrieval cycles
- Designed a self-righting marine buoy using a Blue Robotics pressure vessel, dome cap, & t-slot frame with custom tether mounts & modular ballast weights for reliable above-water cellular transmission
- Collaborated on a dual-servo winch with spur gear speed reduction & ratchet-pawl locking for tether-buoy depth control

GOAL! (Penalty Shootout Arcade Game) | *Advanced Mechatronics I (Graduate)*

- Designed & fabricated a 2-player electromechanical game featuring a 3D-printed rack & pinion kicker & servo goalie
- Programmed an Arduino Mega to control game logic & real-time audio/visual feedback from joystick & sensor inputs
- Integrated a jam-proof optical coin system & organized all power, signal, & actuator wiring into a clean, serviceable layout, supporting uninterrupted operation during a 6-hour public demo

Mobile Manipulator Maze Challenge | *Robotics Final Project*

- Navigated to 3/4 QR codes, successfully picked up & delivered a sock to its target, earning joint-highest course score
- Integrated mobile navigation & arm manipulation using ROS and MoveIt! on Ubuntu, implementing motion planning & visualization in Rviz

Technical Skills

CAD & Simulation: SOLIDWORKS (Certified Associate) · Siemens NX · Ansys FEA/CFD
Programming & Software: MATLAB · ROS · MoveIt! · Simulink · Python · Git · Linux
Hardware & Fabrication: Arduino · 3D Printing (FDM, SLS) · Soldering · Circuit Design · Laser Cutting · GD&T
Independent Study: CS50X · CS50P (In Progress)

Leadership & Activities

- Pi Tau Sigma Mechanical Engineering Honors Society
- Course Grader, Mechatronics: Haptics (Graduate)
- SHPE Community Outreach Director
- Summer Engineering Seminar Mentor